

Scaling Factor 3: Proportion of days that are school days in the ozone season = 0.393. (Ozone is modeled for the 5 months from May 1 through September 30. We assume that children are in school during weekdays for all of May, 2 weeks in June, 1 week in August, and all of September. This corresponds to approximately 2.75 months out of the 5 month season, resulting in an estimate of 39.3% of days ($2.75/5 \times 5/7$).)

G.2.4.2 Gilliland et al. (2001)

See full study description under Gilliland et al. (2001) in Appendix F, Section F.4.1.

G.2.4.3 Ostro and Rothschild (1989)

See full study description under Ostro and Rothschild (1989) in Appendix F, Section F.4.2.

G.2.5 Asthma-Related Effects

Table G-15 summarizes the additional health impacts functions used to estimate the relationship between ozone and asthma exacerbation. Below, we present a brief summary of each of the studies and any items that are unique to the study. Based on advice from the SAB-HES (U.S. EPA-SAB 2004), regardless of the age ranges included in the source epidemiology studies, we extend the applied population to ages 6 to 18, reflecting the common biological basis for the effect in children in the broader age group.

Table G-15. Additional Health Impact Functions for Ozone and Asthma-Related Effects

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Asthma Exacerbation, One or More Symptoms	Mortimer et al.	2002	Eight U.S. urban areas	6-18		D8HourMax	0.009288	0.00387	Logistic	Uses incidence rate of 0.116
Asthma Exacerbation, One or More Symptoms	O'Connor et al.	2008	Seven U.S. urban communities	6-18	PM _{2.5} , NO ₂	D8HourMax	0.0009661	0.002991	Log-linear	Uses incidence rate of 0.207
Asthma Exacerbation, One or More Symptoms	Schildcrout et al.	2006	Eight U.S. urban areas	6-18		D8HourMax	0.00222	0.002822	Logistic	Uses incidence rate of 0.52

G.2.5.1 Mortimer et al. (2002)

Mortimer et al. (2002) examined the effects of daily levels of PM₁₀, SO₂, NO₂ and ozone on peak expiratory flow rate and asthma symptoms among 846 asthmatic children aged four to nine living in eight urban areas in the Northeast and Midwest. Exposures were estimated using EPA's AIRS database and averaging pollutant concentrations at all monitors in each subject's county. Effects were analyzed using mixed linear models and generalized estimating equation models. The authors found an association in single-pollutant models between each pollutant and self-reported morning asthma symptoms.

The health impact function in BenMAP is based on the odds ratio for a 15 ppb increase in 4-day average 8-hour mean ozone between June and August (OR=1.16, 95%CI 1.02-1.30), adjusted to 8-hour max exposure metric as described below.

G.2.5.2 O'Connor et al. (2008)

O'Connor et al. (2008) studied the association between asthma symptoms and daily outdoor air pollutant concentrations (NO₂, SO₂, CO, ozone, and PM_{2.5}) among 937 asthmatic children aged five to 12 in seven inner-city areas in the U.S. Exposure data were estimated from monitor data from EPA's AIRS database for the years 1998 to 2001. Outcome data were collected via telephone interview every two weeks. The effect of pollutant levels on frequency of asthma symptoms was assessed using generalized estimating equation models. The health impact function in BenMAP is based on the relative risk for a 26.7 ppb increase in 19-day average 8-hour mean ozone between June and August (OR=1.04, 95%CI 0.82-1.32), adjusted to the 8-hour max exposure metric as described below.

Incidence Rate: Daily incidence rate for days with wheeze, tightness in chest, cough 21% (Table 1).

G.2.5.3 Schildcrout et al. (2006)

Schildcrout et al. (2006) investigated the relation between ambient concentrations of the five criteria pollutants (PM₁₀, O₃, NO₂, SO₂, and CO) and asthma exacerbations (daily symptoms and use of rescue inhalers) among 990 children in eight North American cities during the 22-month prerandomization phase (November 1993-September 1995) of the Childhood Asthma Management Program. Short-term effects of CO, NO₂, PM₁₀, SO₂, and warm-season O₃ were examined in both one-pollutant and two-pollutant models, using lags of up to 2 days in logistic and Poisson regressions. When modeling ozone, the authors limited the study period to warm months (May through September).

Asthma Exacerbation, One or More Symptoms

In a single-pollutant model, Schildcrout et al. (2006, Figure 1) reported an odds ratio of 1.06 (95% CI: 0.92, 1.23) for daily asthma symptoms associated with 30 ppb change in 24-hr mean of ozone at lag 0.

Incidence Rate: Daily incidence rate for one or more symptoms (symptom score>0) = 52% (Schildcrout, et al., 2006, Table 1).

G.2.6 Converting Functions to 8-Hour Daily Maximum Metric

A number of health impact functions were converted from 1-hour maximum and 24-hour average to the 8-hour maximum metric. To convert, say, a 1-hour maximum function, we multiplied the 1-hour maximum coefficient with the ratio of the typical 1-hour maximum value to the typical 8-hour maximum value from Anderson and Bell (2010), Table 2. We approximated 8-hour average metric as equivalent to the 8-hour maximum metric due to high correlation between the two metrics.

Appendix H. Core Health Valuation Functions in U.S. Setup

This appendix presents the core unit values that are available in BenMAP for each of the health endpoints included in the current suite of health impact functions. Specifically, this appendix includes the values currently used by U.S. EPA in regulatory impact analyses. For the .apvx files summarizing current EPA practices, see:

<https://www.epa.gov/benmap/benmap-community-edition>

Wherever possible, we present a distribution of the unit value, characterizing the uncertainty surrounding any point estimate. The mean of the distribution is taken as the point estimate of the unit value, and the distribution itself is used to characterize the uncertainty surrounding the unit value, which feeds into the uncertainty surrounding the monetary benefits associated with reducing the incidence of the health endpoint. Below we give detailed descriptions of the derivations of unit values and their distributions, as well as tables listing the unit values and their distributions, available for each health endpoint. The definitions of the distributions and their parameters are given in Table H-1.

Table H-1. Unit Value Uncertainty Distributions and Their Parameters

Distribution*	Parameter 1 (P1)	Parameter 2 (P2)
Normal	Standard deviation	–
Triangular	Minimum value	Maximum value
Lognormal **	Mean of corresponding normal distribution	Standard deviation of corresponding normal distribution
Uniform	Minimum value	Maximum value
Weibull ***	α	β

*In all cases, BenMAP calculates the mean of the distribution, which is used as the “point estimate” of the unit value.

** If Y is a normal random variable, and $Y = \log_e X$, then X is lognormally distributed. Equivalently, X is lognormally distributed if $X = e^Y$, where Y is normally distributed.

*** The Weibull distribution has the following probability density function:

$$\left(\frac{\beta}{\alpha}\right)\left(\frac{x}{\alpha}\right)^{\beta-1} e^{-(x/\alpha)^\beta}$$

This appendix also presents EPA methods for developing income growth adjustment factors that allow BenMAP-CE users to adjust the WTP estimates to account for the growth in income over time.

H.1 Mortality

The economics literature concerning the appropriate method for valuing reductions in premature mortality risk is still developing. The adoption of a value for the projected reduction in the risk of premature mortality is the subject of continuing discussion within the economics and public policy analysis communities. Issues such as the

appropriate discount rate and whether there are factors, such as age or the quality of life, that should be taken into consideration when estimating the value of avoided premature mortality are still under discussion. BenMAP currently offers a variety of options reflecting the uncertainty surrounding the unit value for premature mortality.

H.1.1 Value of a Statistical Life Based on 26 Studies

The current undiscounted VSL used by EPA is \$8.7 million (2015\$). This estimate is the mean of a distribution fitted to 26 “value of statistical life” (VSL) estimates that appear in the economics literature and that have been identified in the Section 812 Reports to Congress as “applicable to policy analysis.” This represents an intermediate value from a variety of estimates, and it is a value EPA has frequently used in Regulatory Impact Analyses (RIAs) as well as in the Section 812 Retrospective and Prospective Analyses of the Clean Air Act. Accounting for the cessation lag, or the delay between pollutant exposure and death, this VSL equates to \$8.1 million (2015\$) using a 2% discount rate, \$7.9 million (2015\$) using a 3% discount rate and \$7.1 million (2015\$) using a 7% discount rate (U.S. EPA, 2014).

The VSL approach mirrors that of Viscusi (1992), and uses the same criteria as Viscusi in his review of value-of-life studies. The \$8.7 million estimate is consistent with Viscusi’s conclusion (updated to 2015\$) that “most of the reasonable estimates of the value of life are clustered in the \$5.2 to \$12.3 million range.” Five of the 26 studies are contingent valuation (CV) studies, which directly solicit WTP information from subjects; the rest are wage-risk studies, which base WTP estimates on estimates of the additional compensation demanded in the labor market for riskier jobs. Because this VSL-based unit value does not distinguish among people based on the age at their death or the quality of their lives, it can be applied to all premature deaths. Table H-2 presents the unit values for the 26 value-of-life studies, the 2%, 3% and 7% discounted unit values represent the core EPA values for this endpoint while the undiscounted rate represents an additional valuation function.

Table H-2. Core Unit Values for VSL based on 26-value-of-life studies

Basis for Estimate *	Age Range at Death		Unit Value (VSL) (2015\$)	Distribution of Unit Value	Parameters of Distribution	
	Min	Max			P1	P2
VSL, based on 26 value-of-life studies	0	99	8,705,114	Weibull	9,648,168	1.509588
VSL, based on 26 value-of-life studies, 2% discount rate	0	99	8,132,666	Weibull	9,013,705	1.410317
VSL, based on 26 value-of-life studies, 3% discount rate	0	99	7,887,356	Weibull	8,741,819	1.367777
VSL, based on 26 value-of-life studies, 7% discount rate	0	99	7,103,778	Weibull	7,873,354	1.231894